

Dielectric Slab waveguide

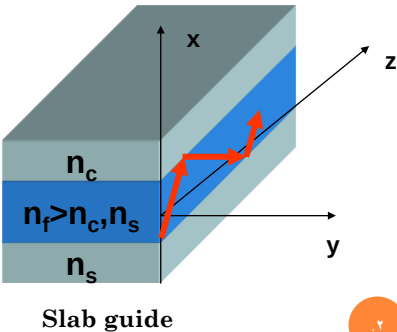
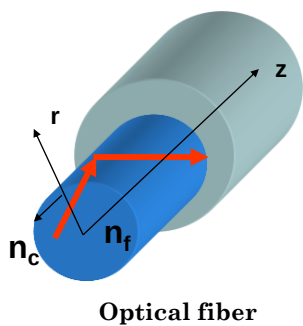
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SLAB WAVEGUIDE

- Open Guide = Dielectric waveguide



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Wave Equation:

Maxwell Equations:

For dielectric waveguide:

$$\text{we have } \begin{matrix} \sigma = 0 \\ \rho = 0 \end{matrix} \rightarrow J = 0$$

$$\rightarrow \begin{matrix} \nabla \wedge H = \delta D / \delta t \\ \nabla \cdot B = 0 \end{matrix} \quad \text{et} \quad \begin{matrix} \nabla \wedge E = - \delta B / \delta t \\ \nabla \cdot D = 0 \end{matrix}$$

• Thus
$$\nabla \wedge (\nabla \wedge E) = - \nabla \wedge (dB/dt) = - d^2(\epsilon\mu E)/dt^2$$

$$\nabla \wedge (\nabla \cdot E) - \nabla^2 E = - \epsilon\mu (d^2E/dt^2)$$

• With ϵ constant $\rightarrow \nabla \cdot D = \nabla \cdot \epsilon E = \epsilon (\nabla \cdot E) = 0$

• Wave Equation:
$$\nabla^2 E = \epsilon\mu (d^2E/dt^2) \quad \text{-----}(A)$$

E: Electric field Intensity
D: Electric Displacement field vector

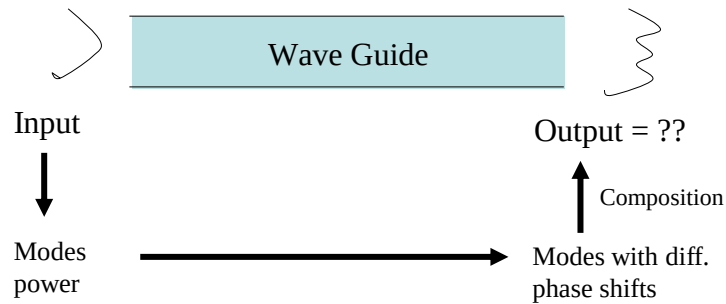
H: Magnetic field Intensity
B: Magnetic flux density

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MODAL ANALYSIS

• why ?



Mode = Field profile $\phi(x,y)$ + Propagation constant β

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MODAL ANALYSIS

*The V number determines the modes supported by the waveguide, and the input field excites some or all of these modes.

*In the modal analysis, we analyze the input field in terms of the waveguide modes to get the weight of each mode.

*Knowing the propagation constant β of each mode, we can calculate the phase shift of each mode after the propagation through the length of the waveguide.

*At the end of the waveguide, we can get the output field by the summation of the different modes at the output, each with its gained phase.

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Propagation Modes

○Harmonic field (format of modal field)

$$E(x,y,z,t) = E(x,y) e^{j(\omega t - \beta z)}$$

Modal Wave Equation

From equ. A $\Rightarrow \nabla_t^2 E(x,y) - (\beta^2 - k_n^2) E(x,y) = 0$

where Refractive index $n = c\sqrt{\mu\epsilon}$

○For 1D waveguide $\Rightarrow$ 2 independent solutions

TE ($E_z=0$)
transverse electric

& TM ($H_z=0$)
transverse magnetic

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EX: STEP INDEX SYMMETRIC WAVEGUIDE (1)

GUIDED MODES

$$\begin{aligned} n(x) &= n_1 & \text{for } |x| < d \\ &= n_2 & \text{for } |x| > d \end{aligned}$$

General Field solution

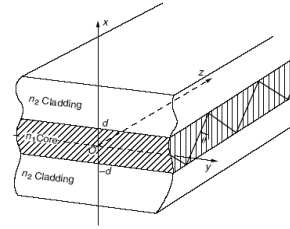
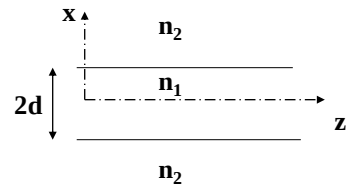
$$E_y = A_I e^{-j h_1 x} + B_I e^{+j h_1 x} \quad |x| < d$$

$$= A_{II} e^{-j h_2 x} + B_{II} e^{+j h_2 x} \quad x > d$$

$$= A_{III} e^{-j h_2 x} + B_{III} e^{+j h_2 x} \quad x < -d$$

$$\begin{aligned} \text{where } h_1^2 &= k_{n1}^2 - \beta^2, \\ h_2^2 &= k_{n2}^2 - \beta^2 = -p^2 \\ k_{n1} &= n_1 k_0 \quad \text{and} \\ k_{n2} &= n_2 k_0 \end{aligned}$$

$$\text{Note that } n_1 > n_{\text{eff}} > n_2 \rightarrow k_{n1} > \beta > k_{n2}$$



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BOUNDARY CONDITIONS

IN OPTICS

Boundary conditions are needed to complete the solution of wave equation

For Perpendicular Electric Field

$$D_{1n} - D_{2n} = 0$$

For Perpendicular Magnetic Field

$$B_{1n} - B_{2n} = 0$$

For Tangential electric Field

$$E_{1t} - E_{2t} = 0$$

While for tangential magnetic field

$$H_{1t} - H_{2t} = 0$$

Note : $D = \epsilon E$, $B = \mu H$



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FIELD POLARIZATION

• We may have two solutions:

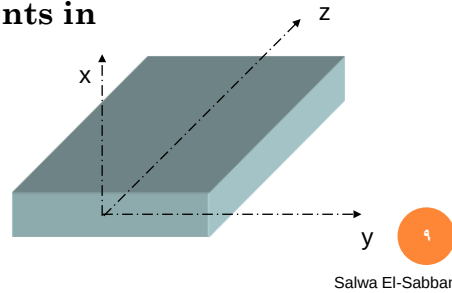
- TE solution with field components

$$E_y, H_x \text{ and } H_z$$

- TM solution with field components

$$H_y, E_x \text{ and } E_z$$

→ Tangential Components in Y and Z directions



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S. I. symmetric waveguide (2)

- Boundary conditions at ($x = \pm \infty$):

$E_y = 0$ when $x = \pm \infty$ for guided modes

$$\rightarrow B_{II} = 0 \quad \& \quad A_{III} = 0$$

- Apply TE Boundary conditions at ($x = \pm d$):

$$- E_{Iy}(d) = E_{IIy}(d) \quad E_{Iy}(-d) = E_{IIIy}(-d)$$

$$- H_{Iz}(d) = H_{IIz}(d) \quad H_{Iz}(-d) = H_{IIIz}(-d)$$

$$\begin{aligned} \bullet E_y &= A_I e^{-jh_1x} + B_I e^{+jh_1x} & |x| < d &\rightarrow C_1 \cos(h_1x) + C_2 \sin(h_1x) \\ &= A_{II} e^{-jh_2x} + B_{II} e^{+jh_2x} & x > d &\rightarrow A_{II} e^{-px} \\ &= A_{III} e^{-jh_2x} + B_{III} e^{+jh_2x} & x < -d &\rightarrow B_{III} e^{px} \end{aligned}$$

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Characteristic Equation

$$\begin{aligned} \rightarrow E_{Iy}(d) - E_{IIy}(d) &= 0 \\ \rightarrow H_{Iz}(d) - H_{IIz}(d) &= 0 \end{aligned}$$

$$\begin{aligned} E_{Iy}(-d) - E_{IIIy}(-d) &= 0 \\ H_{Iz}(-d) - H_{IIIz}(-d) &= 0 \end{aligned}$$

$$\text{Note: } \nabla \Lambda E = -dB/dt = -j\omega\mu H \rightarrow H_z = (j/\omega\mu)\delta E_y/\delta x$$

$$\begin{aligned} H_z &= (-jh_1/\omega\mu)[C_1 \sin(h_1 x) - C_2 \cos(h_1 x)] && \text{in region I} \\ H_z &= (-jp/\omega\mu)A_{II}e^{-px} && \text{in region II} \\ H_z &= (jp/\omega\mu)A_{III}e^{px} && \text{in region III} \end{aligned}$$



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Characteristic Equation

From Boundary conditions

$$C_1 \cos(h_1 d) + C_2 \sin(h_1 d) = A_{II} e^{-pd}$$

$$C_1 \cos(-h_1 d) + C_2 \sin(-h_1 d) = B_{III} e^{-pd}$$

$$(-jh_1/\omega\mu)[C_1 \sin(h_1 d) - C_2 \cos(h_1 d)] = (-jp/\omega\mu)A_{II} e^{-pd}$$

$$(-jh_1/\omega\mu)[-C_1 \sin(h_1 d) - C_2 \cos(h_1 d)] = (jp/\omega\mu)B_{III} e^{-pd}$$

- Put BC in matrix form $[M] \cdot [A] = [0]$



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Characteristic Equation (Dispersion Relation)

From Boundary conditions

$$\begin{bmatrix} \cos(h_1 d) & \sin(h_1 d) & -e^{-pd} & 0 \\ \cos(h_1 d) & -\sin(h_1 d) & 0 & -e^{-pd} \\ (-jh_1/\omega\mu)\sin(h_1 d) & (jh_1/\omega\mu)\cos(h_1 d) & (jp/\omega\mu)e^{-pd} & 0 \\ (jh_1/\omega\mu)\sin(h_1 d) & (jh_1/\omega\mu)\cos(h_1 d) & 0 & (-jp/\omega\mu)e^{-pd} \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \\ A_{II} \\ B_{III} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

- Condition of solution Det. [M] = 0 C/C equation.

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Characteristic Equation (Dispersion Relation)

→ **Dispersion relation for TE modes**

$$(h_1 \tan(h_1 d) - p)(p \tan(h_1 d) + h_1) = 0$$

$$\begin{aligned} & h_1 \tan(h_1 d) = p && \text{(Even modes)} \\ \text{or} & p \tan(h_1 d) = -h_1 && \text{(Odd modes)} \end{aligned}$$

, **Similarly, for TM modes we have:**

$$\begin{aligned} & h_1 \tan(h_1 d) = (n_1/n_2)^2 p && \text{(Even modes)} \\ \text{or} & p \tan(h_1 d) = -(n_2/n_1)^2 h_1 && \text{(Odd modes)} \end{aligned}$$

- Solve dispersion relation to get h_1 , p , β
- Symmetric waveguide → Symmetric mode
(Even symmetric mode or Odd symmetric mode)

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Normalized Parameters

- Normalized frequency V

$$V = 2 k_0 d [n_1^2 - n_2^2]^{1/2}$$

Mode effective index

$$n_{\text{eff}} = \beta/k_0$$

- Normalized propagation constant b

$$b = \frac{n_{\text{eff}}^2 - n_2^2}{n_1^2 - n_2^2}$$

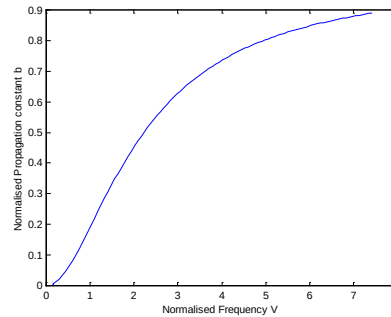
- Normalized c/c equation

$$\tan \left[\frac{V}{2} \sqrt{1-b} \right] = \sqrt{\frac{b}{1-b}}$$

- For single mode $V < \pi$**

- Number of modes

$$N \sim 1 + \text{Int.}(V/\pi)$$



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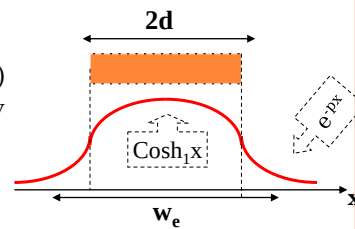
Fundamental TE mode

For TE modes we solve for E_y or H_z

- To get the field profile (field distribution) of the fundamental mode (even mode), apply the boundary condition

→

$$E_y = \begin{cases} E_m \cos h_1 x & |x| < d \\ E_m \cos h_1 d e^{-p[|x|-d]} & |x| > d \end{cases}$$



E_m is the field Max.

- Effective width $w_e = 2d + 2/p$
- Effective width is the mode width when its electric field falls to $1/e$ from its value at the guide width.

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Fundamental TE mode

• Optical power $P_o = \int_{-\infty}^{+\infty} E \wedge H dx = \left(\frac{\beta}{\omega \mu} \right) \int_{-\infty}^{+\infty} |E_y(x)|^2 dx$

$$P_o = \left(\frac{\beta}{\omega \mu} \right) P_n$$

• Thus, Normalized power $P_n = \int_{-\infty}^{+\infty} |E_y(x)|^2 dx = E_m^2 \left(d + \frac{1}{p} \right)$

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Effective Guide Width

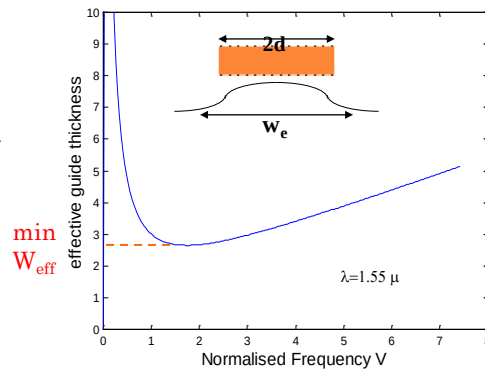
In this figure, the increase of $V_{no.}$ is due to the increase of guide width $2d$

• Effective width W_{eff}
 $W_{eff} = 2d + 2/P$

• Min. Effective width

• Confinement factor

$$\Gamma = \frac{\int_{-d}^d |E|^2 dx}{\int_{-\infty}^{+\infty} |E|^2 dx}$$



Confinement factor determines the ratio between the power of the mode in the region outside the guide to the total mode power

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Effective Guide Width

*In the left section of the previous curve (before the point of minimum W_{eff}), the increase of the V_{no} as a result of the increase of guide width leads to higher β

→ higher $p \rightarrow$ rapid decay for the field outside the waveguide (e^{-px}) → Lower effective width.

*In the right section of the previous curve (after the point of minimum W_{eff}), the increase of the V_{no} leads to the increase of the effective width due to the increase of the guide width itself ($2d$).

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Example

Calculate the effective index of a glass guide of index 1.5 @ $\lambda=1.3 \mu\text{m}$ surrounded by air when the guide thickness is 420 nm.

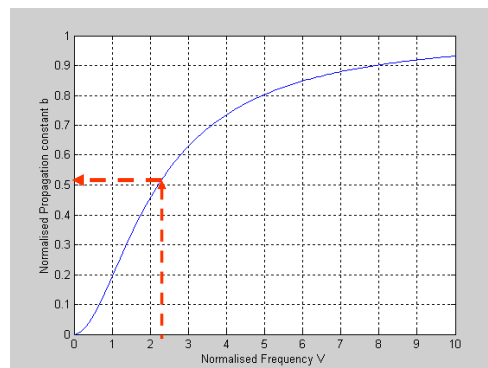
○ We have:

$$\begin{aligned} \bullet K_0 &= 2\pi/\lambda = 4.833 \mu\text{m}^{-1} \\ \bullet V &= 2k_0 d (n_1^2 - n_2^2)^{1/2} \\ &= 2 \times 4.833 \times 0.21 (1.5^2 - 1)^{1/2} \\ &= 2.268 < \pi \end{aligned}$$

→ SM guide

○ Go to V- b graph

$$\begin{aligned} \rightarrow \text{From graph } b &= 0.5 \\ \rightarrow n_{\text{eff}}^2 &= n_2^2 + b(n_1^2 - n_2^2) \\ &= 1 + 0.5(1.5^2 - 1) \\ n_{\text{eff}} &= 1.274 \end{aligned}$$



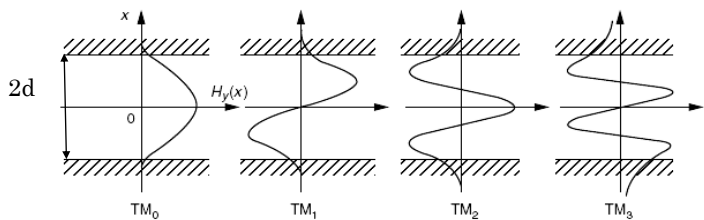
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MODE FIELD PROFILE

Optical Field Distribution of TM Guided Modes



fundamental mode

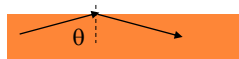
mode order 0 mode order 1 mode order 2 mode order 3

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Mode cut-off

- For symmetric guide
- Fundamental → No cutoff
- Higher order Cut-off → Radiation



Guided mode $\theta > \theta_c$

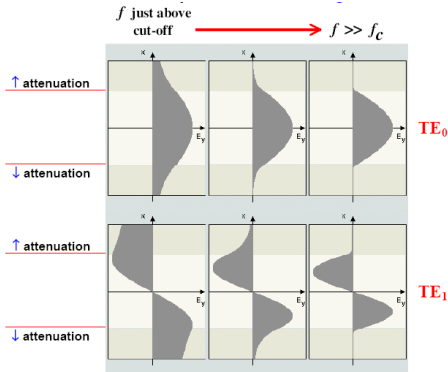


Radiation mode $\theta < \theta_c$

The cut-off V_{no} of the mode with order m is

$$V_{no_{cutoff}} = m\pi$$

Field profile near cut-off



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Cutoff frequency

The cutoff frequency of mode of order m is the minimum frequency of the optical wave incident to the waveguide that allows this mode to exist.

To get the cutoff frequency of mode of order m

$$m\pi = 2 k_0 d [n_1^2 - n_2^2]^{1/2}$$

$$f_c = \frac{mc}{4d\sqrt{n_1^2 - n_2^2}}$$

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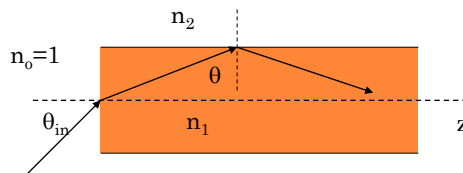
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Guide Numerical Aperture

As in the fiber, NA is related to the Max. angle that an incident ray can have and still be guided

$$n_o \sin \theta_{in-max} = \sqrt{n_1^2 - n_2^2} = NA$$



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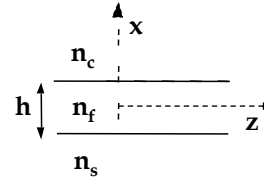
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Ex. Step Index Asymmetric WG (1)

- $n(x) = n_c$ for $x > h/2$
 $= n_f$ for $|x| < h/2$
 $= n_s$ for $x < -h/2$

- Let: $n_f > n_s > n_c$



$$\begin{aligned}
 E_y &= C_1 \cos(h_1 x) + C_2 \sin(h_1 x) & |x| < d &\rightarrow C_3 \cos(h_1 x + \Psi) \\
 &= A_{II} e^{-p x} & x > d \\
 &= B_{III} e^{\alpha x} & x < -d
 \end{aligned}$$

- Find:

- C/c equation
- Field distribution
- Cut-off condition

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Ex. Step Index Asymmetric WG (2)

Define:

Normalized frequency and film thickness

$$V \equiv kh \sqrt{n_f^2 - n_s^2}$$

Normalized guide index

$$b \equiv \frac{N^2 - n_s^2}{n_f^2 - n_s^2}$$

$b = 0$ at cut-off ($N = n_s$), and approaches 1 as $N \rightarrow n_f$.

Measure for the asymmetry

$$a \equiv \frac{n_s^2 - n_c^2}{n_f^2 - n_s^2} \text{ for TE, } \quad a \equiv \frac{n_f^4}{n_c^4} \frac{n_s^2 - n_c^2}{n_f^2 - n_s^2} \text{ for TM}$$

N is the n_{eff} of the mode

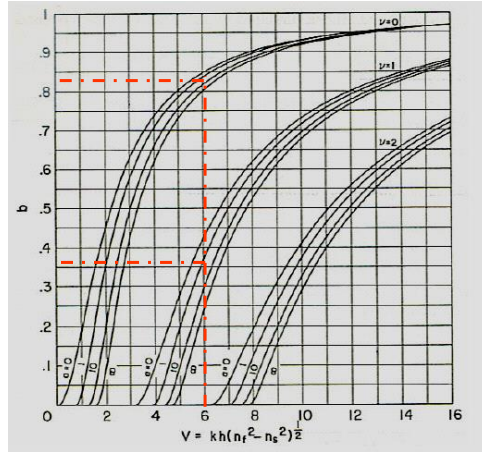
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NORMALIZED DISPERSION CURVES

Calculate V
 Calculate a
 Get $b_1, b_2, \dots$ etc
 Calculate $n_{\text{eff}1}, n_{\text{eff}2}, \dots$ etc



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Modal Analysis(1) :

How ?

Wave equation + Boundary conditions = Propagation modes

Mode = Field distribution $E_v(x)$ + Propagation constant β_v

Field profile

$$E(x, z) = \sum_{v=0}^{m-1} c_v \psi_v(x) e^{-j\beta_v z}.$$

c_v is the mode amplitude

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Modal Analysis (2)

- Input Field Profile

$$E(x,0) = \sum_{v=0}^{m-1} c_v \psi_v(x)$$

For m guided modes

- Orthogonality

$$\int_{-\infty}^{+\infty} \psi_n(x) \psi_m^*(x) dx = 0 \quad \text{for} \quad m \neq n$$

Modes are orthogonal

- Give:

$$c_v = \frac{\int_{-\infty}^{+\infty} E(x,0) \psi_v^*(x) dx}{\int_{-\infty}^{+\infty} |\psi(x)|^2 dx}$$



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